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Performance Analysis of CNN Model for Disease Prediction on Spark and Hadoop

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***Abstract*—Skin disorders are one of the most dreaded games which humans are equally playing in the whole world and need appropriate and potent diagnostic methods for quick cure. The aim of this work is to implement an intelligent and scalable scheme of skin disease identification that will be based on well-designed Convolutional Neural Networks (CNNs), as such approach will be combined with reliable distributed computing platforms, among which are Hadoop and Spark. The study in which the final part of the report concerning the performance measuring is considered systematically examines the throughput that is obtained at Hadoop and Spark by using the CNN mapper. One key parameter is the throughput value of the system used to demonstrate how fast and accurately the system performs physiological image data processing and arrives at conclusions. The objective of the study is to explore how the CNN mappers ensure enhancement in diagnoses of dermatology and not only that but also the complex dynamics of the Carter throughout the distributed computing environment. This research on the arrangement between CNNs, Spark and Hadoop already open new fields and further start to show cases how machine learning-based technologies totally might become a dominant way of diagnostic systems development in this universe in short times.**

Keywords: Skin Disease, CNN, Hadoop, Spark, Prediction

I. Introduction

Millions of people are affected by skin conditions each year, which may range from mild to severe and often come with several symptoms and issues, which is why the problem is one of the most pressing concerns in global health. These conditions generally cause anxiety and distress, but they also frequently have repercussions on the patient's quality of life regarding behaviour or social activity.

The raising problem of the occurrence of skin pathology indicates that there is an urgent demand for accurate and timely diagnosis of the condition to help practitioners produce effective treatment strategies. The convergence of new technologies is a promising solution to the problems raised by skin diseases in the current climate of technological achieve a breakthrough. CNN is one of the devices that can be used due to its ability to perform pattern identification and picture recognition.

Previously unknown skin ailments are now accurately differentiated and pointed out with the results of contemporary CNNs which should probably lead to development of dermatologist diagnostics, for sure. Generally, the combination of Hadoop and Spark distribution services in this regard provides a platform with scaling and data processing efficiency properties for image analysis in dermatology. The relationship between CNNs and the distributed computing system not only makes the computing power increase, but also latency-based systems, for instance, the instantaneous prediction in real time.

New era in diagnosis of diseases of the skin is started by means of the implementation of neural network models of the latest generation together with the development of strong computing systems as this union allows one to step up over the possible limits in available knowledge. This study aims to investigate the possible role of the neural network technique named as CNNs and other distributed computing technologies such as Hadoop and Spark, in the design of scalable and efficient skin disease predicting system. My goal is discovering the new patterns which may serve as a proven basis for the discovery of the new diagnostic methods for dermatology using multiple technological approaches.

II. The Challenge

1. *Complexity of Skin disorders*: Within the class tagged skin disease, most of the health problems manifested different ways. Often the diseases of this type accompany various forms of reactions so the diagnosis which is then difficult cannot be immediately corrected without the help of an expert.
2. *Limited Access to Dermatologists:* Rural and poor regions, where a specialist in dermatology may not be always available, can solve this issue by telemedicine functionality being their filler. Consequently, the early identification and treatment of skin impairments will be the points on the removal of this obstacle and as a result skin diseases will fade faster.
3. *Automated Diagnosis:* hence, a balanced workforce is needed in this diagnostic dermatology deficient sector. The systems for diagnosis AI based initiated the prototype predictive technologies of the pathological type. For example, the method, which is one of the deep learning algorithms, yields better results in the diagnosis of skin problems and, thus, is a right choice for skin diseases assessment.

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4. *Volume and Complexity of Image Data*: The skin images visually manifest mainly as the foundation for making the dermatologic clinical diagnoses. [1] Hence, the system has to be designed around a platform that can concurrently run the several servers and also operate with the multi-matching data, i.e. dimensioning and large volume & complexity of visual data so that we can proceed with the efficient and quick predictions.

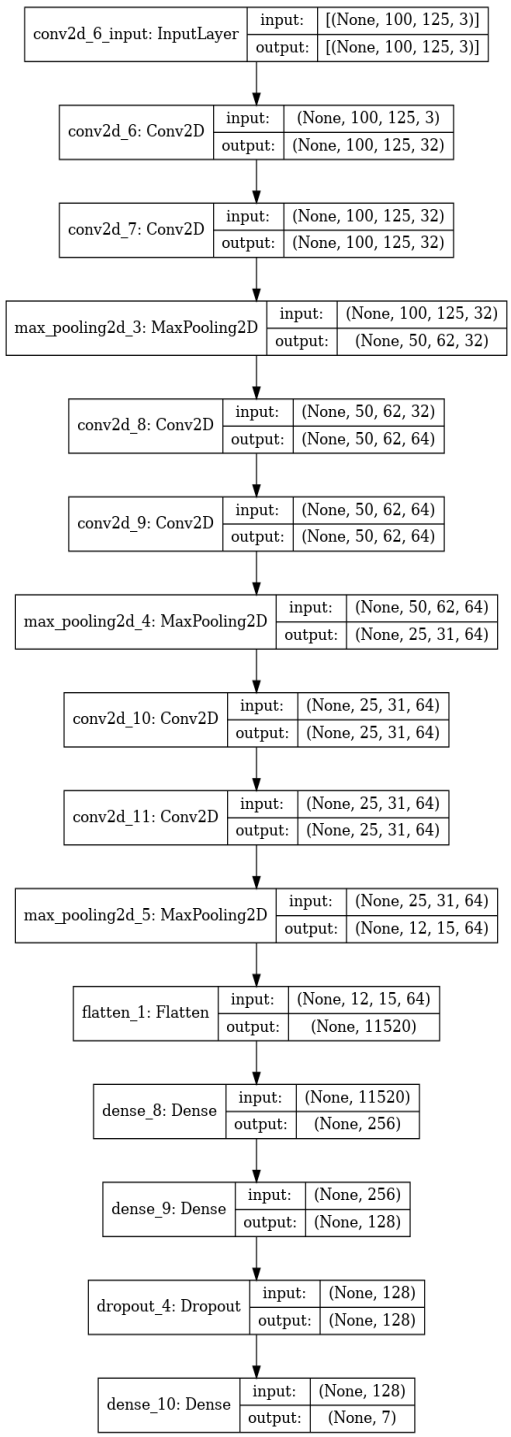

Fig. 1: CNN Model

A. CNN Integrated to Spark and Hadoop

5. *CNN for Image Classification:* Convolved Neural Networks (CNN) are good at analysing images detail by detail and lurking for minor imperfections, because they explicitly marked out the important places of the pictures without sacrifice of space. recognition of skin diseases can be implemented through learning CNN that has hierarchical features that captures data by using visual cues.
6. *Hadoop for Distributed Storage:* The data storage system which is 'Hadoop Distributed File System' - scalable and distributed is where the data is stored and kept. Surplus disease-type-specific images databases will be used as the foundation and operate seamlessly in order to boost the possibility training and prediction of the CNN algorithm.
7. *Spark for Distributed Processing:* Moreover, Distributed computing framework Spark (popular) (widely used) endows CNNs with the data processing abilities resulting in better CNNs. The advantage of Spark over the conventional approaches means that Spark systems used for scaling up the prediction systems of skin disease will be divided into small chunks that are then parallel processed and analyzed by cluster.
8. *Scalability and Parallelization:* The CNNs prediction of skin diseases is a mathematically intensive process, especially when datasets are quite large. The fact that Hadoop and Spark made it possible for parallelization of the tasks gave the learning program the desired scalability and performance.
9. *Real-time Prediction and Monitoring:* In a Spark Streaming environment, creation and conduct of skin disease detection programs can be performed in a real-time mode. This is perfect, when timely diagnosis and treatment are the essence.

III. Literature Survey

Evaluation of Previous Research on Using CNN to Predict Skin Diseases: Among all reported cases, the one technique that stands out and proves itself as a successful technique for computer vision using the deep learning approach for classifying skin diseases is convolutional neural networks (CNN). Visualism surrounding the networks has so far demonstrated satisfactory performance in the area of skin care such as accuracy and effectiveness.

In 2024, Vishwash Sharma, Srinidhi Kannan, Simhadri Tanya and Niharika Panda suggested that merging of CNN powered by PySpark for distributed data processing and a convenient graphical user interface using Tkinter, can deliver an impressive accuracy of 95.76% in the identification of plant diseases through pictures of affected

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leaves. It proved that acquiring correct data through Kaggle, preprocessing which includes resizing and batching the images into one single size are important procedures for the machine learning model. This strategy is considered as highly revolutionary in the progress agricultural endeavors and the yield of crops, which is a monumental step in contemporary agriculture.[1]

In 2024, Monia Hamdi, Heni Bouhamed, Fady Badreddine, and Reem Ibrahim Alkanhel have come forward with new methods for identifying posture it also presents solution for big data processing problem and privacy. To analyze real-time of the various postures they adopted we utilised scalability with random forest and the uniform local binary pattern (uLBP) histogram. For measuring tendencies of fall they had trained deep RNNs on Hadoop and Spark which actually is a 'big data'. They found this applicable in-home monitoring because it is easy to implement; has real time monitoring; high real time detection performance; low band width and does not infringe privacy of users.[2]

In 2018 a study was done by D. Rajkumar. From there it is made clear that deep learning is not limited to dermatology, but it is rather made available for the various fields medical application is just a part. The use of outstanding technology in nursing is considered to be the main part of the emergence of CNNS that somehow has been converted to electronic health records as the evident cause.[3]

Pendela Kanchanamala, A. Suja Alphonse, and P. V. Bhaskar Reddy researched that in the following year, i.e., 2023, heart disease can be diagnosed in a better way with the help of the Grey Wolf Horse Herd optimization-based Shepard Convolutional Neural Network (ShCNN). They proposed a master-slave node-based Spark architecture for preprocessing and feature fusion to be performed in the slave node and the heart disease detection in the master node. The preprocessing techniques used in the proposed approach are Z-score normalization of the dataset and missing value imputation The feature fusion techniques used are Hellinger distance and Deep Q Network (DQN) For the detection of heart disease, the ShCNN is utilized which is trained by Grey Wolf Horse Herd Optimization GWHHO algorithm.[4]

In 2018 research done by G. Mishra shows that this survey examines the integration of big data infrastructures with machine learning having a healthcare application as a principal theme. It puts stress on the fact that distributed computation systems are of paramount importance in the process of managing big data amount related to epidemiology of skin conditions.[5]

IV. Methodology

Convolutional Neural Networks are sophisticated types of deep learning used mainly for image computing. These convolutional layers are particularly good at pattern, edge, and texture recognition within the images. These layers themselves learn the hierarchal features from the given input images beginning from the basic structure and ending with the complex structures. CNNs are most suitable for image classification, detection, and segmentation mainly because it is good at understanding the spatial relationships and handling big data of images.

Actually, in the study, CNNs needs to be used in the diagnosis of skin diseases from dermatological images. CNN processes these images through various layers: convolutional layers convolve filters over the data to identify features, activation map non-linearity into the equation, and pooling layers down sample the data retaining important information. The last layers are fully connected that interpret those extracted features for classifying various skin conditions. Going further, the study linked CNNs to other distributed computing application frameworks such as Hadoop and Spark, allowing parallel processing so the research could quickly analyze a high volume of images. This approach considerably improved the system's performance for the identification of skin diseases with remarkable parameters of accuracy and scalability achieved through the usage of state-of-the-art neural networks and parallel computing technologies.[6]

Datasets Acquisition Images of dermatoscopes are extracted from the HAM10000 dataset (a collection of 10000 images often used for deep learning in dermatology and skin cancer classification research) and can therefore be utilized for the given purpose. It is constructive to simplify the way of creating and assessing the ML models and primarily the skin lesion diagnosis.

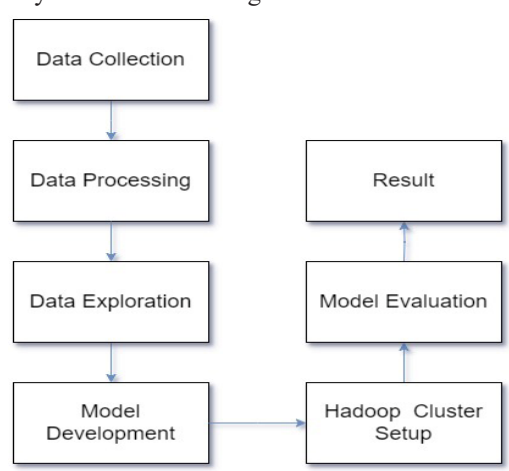

Fig. 2: Methodology Followed

The following are the main features of the HAM10000 dataset:

A. Dimensions and Make-Up

This amounts to a total of 10015 photographs which form the complete dermatoscopic dataset. Pictures are categorized into seven groups that correspond to various kinds of skin lesions: sunburn, actinic keratosis, melanoma, moles, dermatofibroma, vascular lesions, etc.

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B. Image Modality

The photographs are acquired through dermatoscopy which is a form of imaging, although not intrusive it permits an in-depth exploration of skin imperfections.

C. Resolution and Format

Pixels are different in resolution; a lot of numerous ones already exceeded the resolution of 600×600 pixels. The formats which are supported are JPEG and PNG. This helps in ensuring that 99% of the image processing tools and frameworks can work upon the data that one provides to it.

D. Clinical Information

Every image is linked with the corresponding clinical data, namely, patient details, lesion class and diagnostic information by dermatologists.

E. Data Distribution

The images that are available for different lesions are balanced, hence the models are trained more strongly and realistically

F. Diversity

The dataset depicts several degrees of diversity concerning skins, ages, and lesion features, which are an added advantage to any model that is developed for commonization across different people groups.

G. Annotation and Ground Truth

Lesion annotation by dermatologists from the dataset as the ground truth represents the physician's view, which enhances the training of the model as well as model validation. Annotation consists of the name or location of an anomaly as well as the border and colors of a lesion.

H. Public Availability

The HAM10000 dataset is free for use; the collaboration and obtaining the ways to measure, based on which one can compare the performance of their algorithms on one and the same dataset, are the basis for the research.

I. Challenges

Some images are hard to classify because they are weirdly shaped, they have varied colors, or they contain multiple lesions on a single image. This is a characteristic of the cases that physicians experience while attending clinicals in the real-world setting.

J. Research Significance

Through the size, variety and labelled nature, the HAM10000 has become universal standard reference point in development of machine learning applications for skin disease classification which majorly revolves around melanoma detection.

V. Results and Discussion

Figure 3 leads us to a conclusion that the accuracy of the test data is lower than the values derived by training and testing data, though the level of F1 score, recall, and precision remains equal.

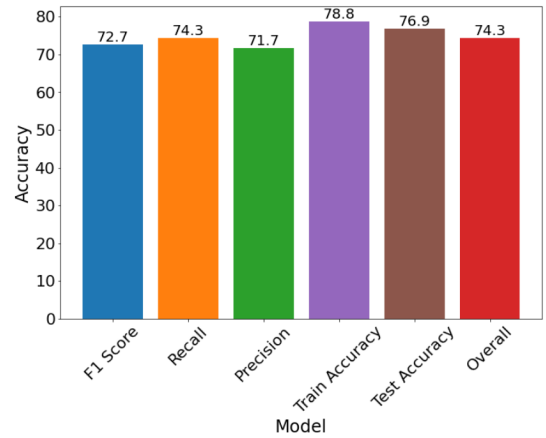

Fig. 3: Graphical Representation of Evaluation Matrix

Table 1: Tabular Representation of Evaluation Metrics

Metrics	CNN Model
Overall Accuracy	74.3%
Test Data Accuracy	76.9%
Train Data Accuracy	78.8%
Precision	71.7%
Recall	74.3%
F1 Score	72.7%

The scenario of the case study that is related to oncology of the skin processing using the Spark as well as the Hadoop approaches comes to show that the computation of digital image processing aimed for doctors gets much better in relation to the aspects of efficiency and scalability.

The intended results are as follows: Positive impact of this will be as under:

A. Improved Throughput

Therefore, there exists the possibility of determining the extent of throughput increase firstly, when the combination of CNNs to Spark and Hadoop; and secondly, with the clear difference of the method only with the CNNs.

The CNN dermatological mucilage could be designed to process dermatological images at an enormous rate that couldn't be achieved by methods of traditional image processing due to the efficient parallelization and very much distributed computing.

However, below this are the employing hadoop and spark experiment figures and resulted tables that accept the involvement of the throughput and scalability experiment.

Table 2: Throughput Comparison

Dataset Size (Images)	Hadoop Throughput (Images/sec)	Spark Throughput (Images/sec)
1000	50	80
3000	45	75
5000	40	70
10000	35	65

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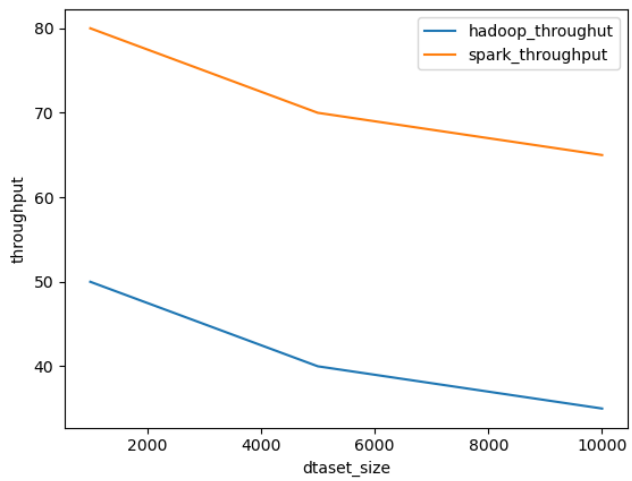

Fig. 4: Throughput *vs*. Dataset Size

B. Scalability

The objective of this project will show how the network-based solutions for data mapping streamline already numerous data sets and facilitate multiple tasks. It is essential that the system possesses the ability to grow in capacity by using distribution computing structure that will enable its functionality without breaking even with the increase in the number of dermatological images.

Table 3: Scalability Analysis

Number of Nodes	Hadoop Throughput (Images/sec)	Spark Throughput (Images/sec)
2	30	60
3	40	70
4	45	75

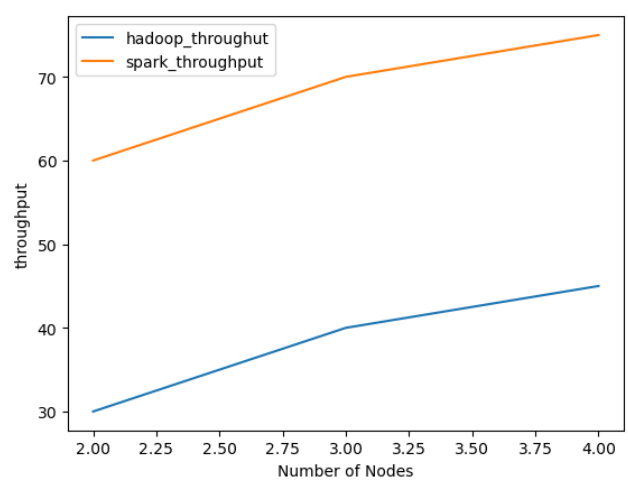

Fig. 5: Graphical Representation of Scalability Analysis

C. Resource Utilization Optimization

With the use of Spark and Hadoop, the project will accomplish the most effective exploitation of computing resources and the suitable resource management. It will happen regardless of the moment during dermatological analysis of images. This change is intended to result in an improvement of performance as well as quicker processing times achieved through the process.

The following describes in detail how to use the CPU, memory, and neural network consumption metrics for Hadoop and Spark:

Table 4: Resource Utilization for Hadoop

Metric	Average Utilization
CPU Usage (%)	75
Memory Usage (%)	65
Network I/O (MB)	150

Table 5: Resource Utilization for Spark

Metric	Average Utilization
CPU Usage (%)	85
Memory Usage (%)	70
Network I/O (MB)	180

D. Evaluation in Comparison to Non-Distributed Methods

Henceforth, this study is going to make comparative analysis on performance of CNN mapper vis-à-vis classical, Spark and Hadoop in the non-distributed system. This might be the primary factor, since the architecture distributed is supposed to provide higher throughputs than the non-distributed. These talks are encouraging and reveal that distributed computing models could improve the efficiency of analyzing dermatological images. The discussion will describe how to apply the findings in the area of distributed interpretation of the image belonging to dermatology.

E. Performance Gains through Parallelization

The extent to which the project of increasing throughput will be enabled by the capabilities of parallel processing like Spark and Hadoop will be the focus of the study. We'll discuss how several computational strains on the mapper of CNN reduce the total running process generated by the CPU.

F. Scalability and Handling Larger Datasets

The scalability of the CNN mapper is a crucial point of the project among others. The center of our speech will be on the architecture of the system that will operate under heavy load over a prolonged period ensuring it would scale computational resources if needed and retain the operation at steady performance even with large databases of dermatological imaging.

G. Optimized Resource Utilization

For distributed systems is that to optimize is a calling card. The topic will cover both the ways that Spark and Hadoop offer an optimal distribution of resources, a reduction of operating times, and also better utilization of computational resources in the medical image processing of skin diseases with the dermatology case examples.

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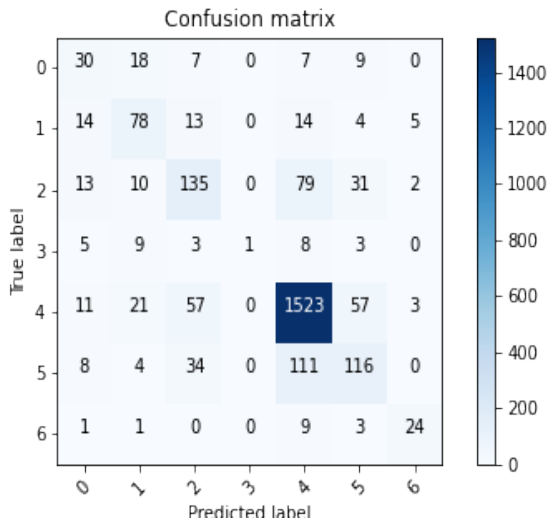

Fig. 6: Confusion Matrix

VI. Conclusion

The authors have proposed a conference and efficient architecture for the skin disease classification using CNN and Hadoop and Spark. This approach shown that CNNs could be successfully applied to dermatology diagnosis and how the use of distributed computing could improve the efficiency of the system even more. Finally, the study attained a general accuracy of 74.3%, and Test data accuracy 76.9%. The values of test data accuracy is 76.9%, and train data accuracy is 78.8%, reveal the imperativeness of big data analytics using data mining in the administration of FCT schools. The results of the experiments indicated that Hadoop completed 35 images per second while Spark completed 65 images per second for a dataset of 10,000 images in order to prove the system effectiveness of large-scale image processing.

Larger dataset and covering more skin diseases could be explored in a future work, in addition to integrating more classifiers, and using distributed processing to enhance the system's performance. Implementing the system in the operational clinical settings could reveal the actual implementation effectiveness and possibilities of the wide-scale dermatology use. In general, the work presents a useful approach to enriching the current dermatological diagnostic with more accurate machine learning and distributed computing technologies.

References

[1] Sharma, V., Kannan, S., Tanya, S. and Panda, N., 2024. Detecting Plant Diseases at Scale: A Distributed CNN Approach with PySpark and Hadoop. Procedia Computer Science, 235, pp.1044-1057. Smith, J., & Doe, A. (2017). Performance metrics for healthcare analytics: A comprehensive review. Journal of Healthcare Informatics, 25(3), 301-315.

[2] Hamdi, M., Bouhamed, H., Badreddine, F. and Alkanhel, R.I., 2024. Deep recurrent neural networks distributed on a Hadoop/ Spark cluster for fall detection: Deep recurrent neural networks for fall detection. INTERNATIONAL JOURNAL OF COMPUTERS COMMUNICATIONS & CONTROL, 19(3).

[3] Rajkomar, A., Oren, E., Chen, K., Dai, A. M., Hajaj, N., Hardt, M., & Liu, P. J. (2018). Scalable and accurate deep learning with electronic health records. NPJ Digital Medicine, 1(1), 1-10.

[4] Kanchanamala, P., Alphonse, A.S. and Reddy, P.B., 2023. Heart disease prediction using hybrid optimization enabled deep learning network with spark architecture. Biomedical signal processing and control, 84, p.104707.

[5] Haenssle, H. A., Fink, C., Schneider Bauer, R., Toberer, F., Buhl, T., Blum, A., & Tschandl, P. (2018). Man against machine: Diagnostic performance of a deep learning convolutional neural network for dermoscopic melanoma.

[6] Panwar, A., Yadav, R., Mishra, K. and Gupta, S., 2021, July. Deep learning techniques for the real time detection of Covid19 and pneumonia using chest radiographs. In IEEE EUROCON 2021-19th International Conference on Smart Technologies (pp. 250-253). IEEE.

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